COM3031 Probabilistic Machine Learning: Week 5

Term 2, 2025-2026

Dr Zeyu Fu, Lecturer in Computer Vision and Machine Learning

z.fu@exeter.ac.uk

In this week

Lectures:

- From Optimisation (VI) to Sampling (MCMC)
- Simple Sampling Methods
- Core MCMC algorithms

Workshop:

- Laplace approximation

Acknowledgements: some lecture slides were adapted from Frank Wood, Mark Schmidt, Michael Gutmann, Willie Neiswanger and Dmitry Vetrov.

Part I: From Optimisation (VI) to Sampling (MCMC)

What do we actually need from the posterior?

Bayesian posterior (from previous weeks)

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}. \quad p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)\, p(\theta)\, d\theta$$

What do we actually need from the posterior?

- In practice, we rarely need the full analytic form of $p(\theta \mid \mathcal{D})$.
- Instead, we usually want to compute expectations, such as:

$$E_{p(\theta|\mathcal{D})}[f(\theta)] = \int f(\theta)p(\theta \mid \mathcal{D})\; d\theta$$

  - These integrals are generally intractable.
- **Examples:** Posterior mean and variance, credible intervals, predictive distributions and model comparison or decision-making quantities.

Variational inference: One Way to Get This

Variational inference (VI)

- Approximate the posterior with a simpler distribution:

$$p(\theta \mid \mathcal{D}) \approx q(\theta), \quad q(\theta) \in \mathcal{Q}$$

- Fit q by minimising reverse KL (equivalently maximising ELBO):

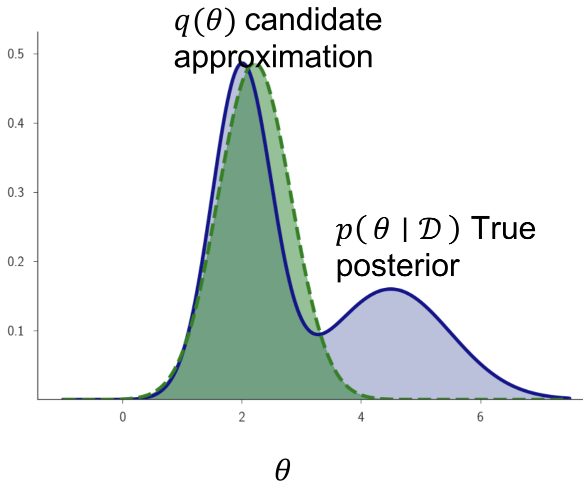
$$q^* = \arg\min_{\mathrm{q}} \mathrm{KL}\left(q(\theta) \,||\, p(\theta \mid \mathcal{D})\right)$$

- Compute expectations under $q(\theta)$, which is tractable.

What VI gives us: fast, deterministic optimisation, scales well to large datasets and often works well in practice

Why VI is not always enough:

- Depends heavily on the chosen family
- Can miss modes and underestimate uncertainty



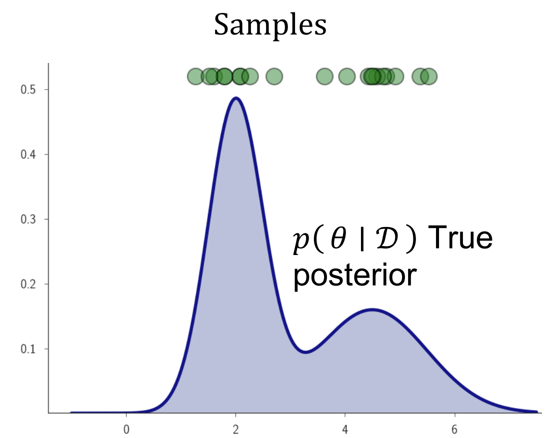
Why Sampling-based Methods

Idea: Instead of approximating the *distribution*, sampling-based methods approximate expectations under the posterior directly

Sampling from the posterior:

- Assume we can draw samples: $\theta^{(1)}, \ldots, \theta^{(N)} \sim p(\theta \mid \mathcal{D})$
- Then for any function $f(\theta)$,

$$E_{p(\theta|\mathcal{D})}[f(\theta)])] \approx \frac{1}{\mathrm{N}} \sum_{\mathrm{i}=1}^{\mathrm{N}} f(\theta^{(\mathrm{i})})$$



Monte Carlo integration

- Approximating expectations by sample averages

Interpretation:

- No assumptions about posterior shape (not Gaussian, not unimodal)
- Naturally captures **multimodality** and **parameter correlations**
- Accuracy improves as the number of samples increases

Monte Carlo Integration

Key posterior-specific examples

- Estimating the posterior mean:

$$\text{If } f(\theta) = \theta,$$

$$\text{Then } E_{p(\theta|\mathcal{D})}[\theta] \approx \frac{1}{\mathrm{N}}\sum_{\mathrm{i}=1}^{\mathrm{N}} \theta^{(\mathrm{i})}$$

- Estimating the posterior variance:

$$\text{If } f(\theta) = (\theta - \bar{\theta})^2,$$

→ uncertainty comes directly from sample spread. $\text{Then } \mathrm{Var}(\theta \mid \mathcal{D}) \approx \frac{1}{\mathrm{N}}\sum_{\mathrm{i}=1}^{\mathrm{N}} (\theta^{(\mathrm{i})} - \bar{\theta})^2$

Why this matters

- We never need the normalising constant of $p(\theta \mid \mathcal{D})$
- We don't need a closed-form posterior
- Sampling turns **Bayesian inference (integral)** into **averaging**

Part II: Simple Sampling Methods

Rejection Sampling

Goal: draw samples from the posterior $p(\theta \mid \mathcal{D})$

- The posterior is known *up to a normalising constant*:

$$p(\theta \mid \mathcal{D}) = \frac{1}{Z}\tilde{p}(\theta), \qquad \tilde{p}(\theta) = p(\mathcal{D} \mid \theta)p(\theta) \quad Z = \int \tilde{p}(\theta)d\theta$$

- We cannot computer Z(the evidence)
- We can evaluate $\tilde{p}(\theta)$which is the unnormalised posterior

Envelope condition

Choose a proposal distribution $q(\theta)$ and a constant M such that

$$\tilde{p}(\theta) \leq \; Mq(\theta) \quad \text{for all } \theta$$

- $Mq(\theta)$ is an envelope/comparison function that sits above the unnormalised posterior

Acceptance probability: sampling $\theta^* \sim q(\theta)$, accept it with probability $\frac{\tilde{p}(\theta^*)}{Mq(\theta^*)} \leq 1$ and the unknown constant Z cancels out.

Rejection Sampling

Application to Bayesian inference:

Substituting: $\tilde{p}(\theta) = p(\mathcal{D} \mid \theta)p(\theta)$,

$$\frac{\tilde{p}(\theta)}{Mq(\theta)} = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{\max_\theta p(\mathcal{D} \mid \theta)p(\theta)}$$

- The acceptance probability is the posterior density relative to its maximum
- High posterior regions are accepted often
- Low posterior regions are rejected

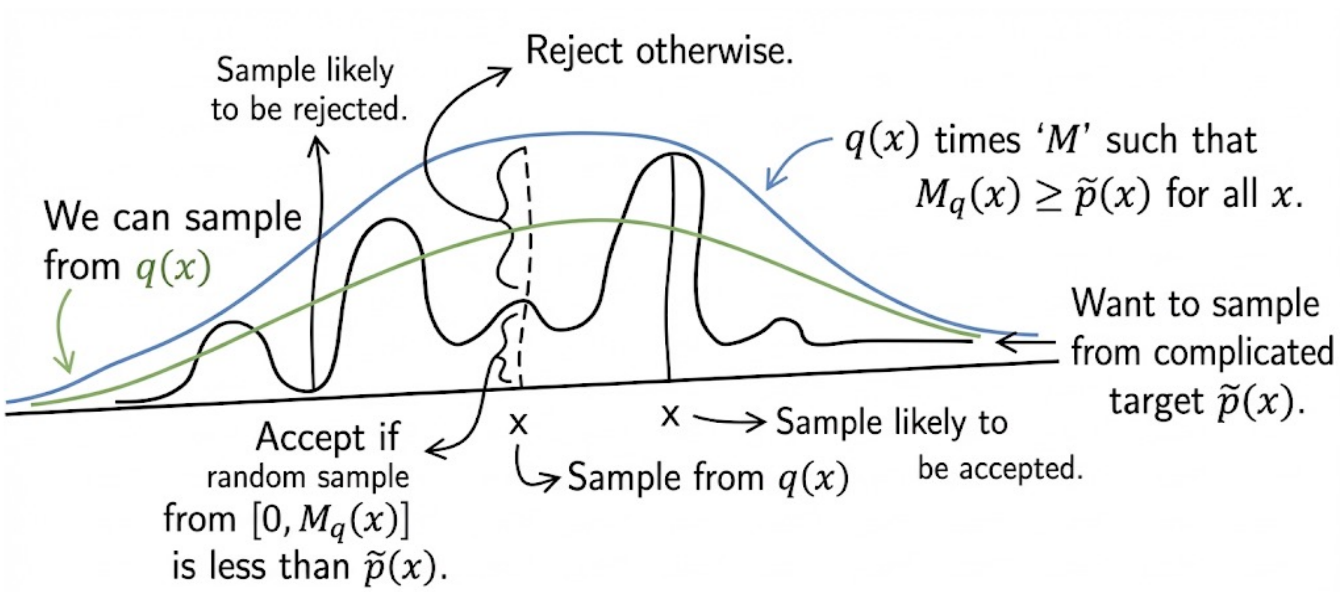
Acceptance test:

- Draw u from a uniform distribution over $\mathrm{Uniform}(0,1)$.
- If $u \leq \frac{\tilde{p}(\theta^*)}{Mq(\theta^*)}$ then the sample θ^* is accepted, otherwise, reject and resample

Rejection Sampling

Why it works

- Accepted samples are uniformly distributed under the curve of $\tilde{p}(\theta)$.
- After normalisation, the accepted θ values follow: $p(\theta) = \frac{1}{Z}\tilde{p}(\theta)$,
- Importantly, the unknown normalising constant Z never needs to be computed



Black: $\tilde{p}(x)$ (unnormalised target)
Green: $q(x)$ (proposal)
Blue: $Mq(x)$ (envelope), with note $Mq(x) \geq \tilde{p}(x)$

Rejection Sampling Algorithm

Given: $\tilde{p}(\theta) = p(\mathcal{D} \mid \theta)p(\theta)$, proposal $q(\theta)$ and envelope constant M

Algorithm:

- Sample θ^* from $q(\theta)$
- Sample u from $\mathrm{Uniform}(0,1)$
- Accept the sample θ^* if $u \leq \frac{\tilde{p}(\theta^*)}{Mq(\theta^*)}$, otherwise reject and repeat

Rejection sampling turns the problem of sampling from a complex distribution into a simple accept–reject test using only the unnormalised density.

Limitations:

- Efficiency depends heavily on how tight the envelope is
- In high dimensions, finding a good $q(\theta)$ and M is very hard
- Acceptance rate can be extremely low

Importance Sampling

From Rejection Sampling to Importance Sampling

- Rejection sampling gives **exact samples** from $p(\theta \mid \mathcal{D}) \propto \tilde{p}(\theta)$, but it is often **inefficient**.
- Instead of rejecting most samples, **importance sampling keeps all draws**, but assigns **weights**.

Target: **posterior**

$$p(\theta \mid \mathcal{D}) = \frac{1}{Z}\tilde{p}(\theta), \qquad \tilde{p}(\theta) = p(\mathcal{D} \mid \theta)p(\theta) \quad Z = \int \tilde{p}(\theta)d\theta$$

- **Goal**: compute posterior expectations (not necessarily draw exact samples):

$$E_{p(\theta|\mathcal{D})}[f(\theta)] = \int f(\theta)p(\theta \mid \mathcal{D}) \; d\theta$$

- **Key trick**: draw sample θ from an easy proposal $q(\theta)$, but not directly from $p(\theta \mid \mathcal{D})$.

$$E_p[f(\theta)] = \int f(\theta)\frac{\tilde{p}(\theta)}{Z}d\theta = \frac{1}{Z}\int f(\theta)\frac{\tilde{p}(\theta)}{q(\theta)}q(\theta), d\theta$$

- **Define unnormalised importance weights:**

$$w(\theta) = \frac{\tilde{p}(\theta)}{q(\theta)}$$

Importance Sampling: Self-Normalised Estimator

Unnormalised importance sampling estimator

- Draw $\theta^{(i)} \sim q(\theta), i = 1, \dots, N$. Then:

$$\widehat{E_p}[f(\theta)] \approx \frac{1}{Z}\frac{1}{N}\sum_{i=1}^{N} f\left(\theta^{(i)}\right)w\left(\theta^{(i)}\right),$$

- Problem: Z is unknown

Self-Normalised Importance Sampling

- Estimate Z using the same samples:

$$Z = \int \tilde{p}(\theta)d\theta = \int \frac{\tilde{p}(\theta)}{q(\theta)} q(\theta)d\theta = E_{\mathrm{q}}[w(\theta)].$$

$$\hat{Z} = \frac{1}{N}\sum_{i=1}^{N} w\left(\theta^{(i)}\right)$$

- Plug in $\hat{Z} \Rightarrow Z$ cancels,

$$E_p[f(\theta)] \approx \frac{\sum_{i=1}^{N} f\left(\theta^{(i)}\right)w\left(\theta^{(i)}\right)}{\sum_{i=1}^{N} w\left(\theta^{(i)}\right)}$$

When Importance Sampling Works

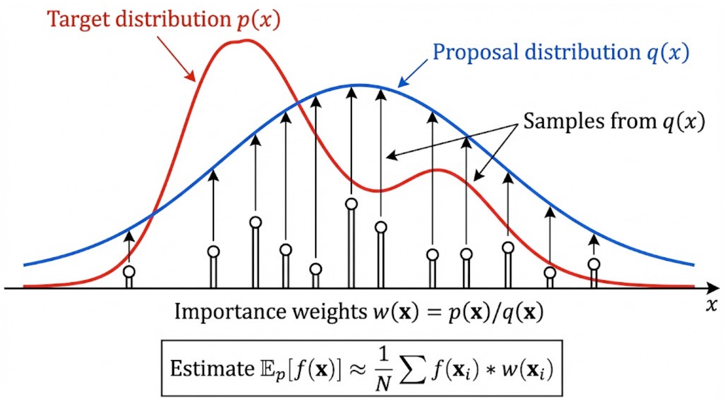
Equivalent "normalised weights" (optional)

Let $w^{(i)} := w\left(\theta^{(i)}\right)$ and $\overline{w}^{(i)} = \frac{w^{(i)}}{\sum_{j=1}^{N} w^{(j)}}$ (normalised weights).

Then, $\widehat{E_p}[f(\theta)] = \sum_{i=1}^{N} \overline{w}^{(i)} f\left(\theta^{(i)}\right)$

- **Key condition (support matching):** $q(\theta) > 0$ wherever $\tilde{p}(\theta) > 0$

  Otherwise: important regions are never sampled.



Efficiency depends on the proposal

- Importance sampling works well if $q(\theta) \approx p(\theta \mid \mathcal{D})$
- Fails when mismatch causes weight degeneracy: most weights ≈ 0 and a few weights dominate → high variance

Common failure: lighter tailed proposal

- If q has lighter tails than p: tail samples are rare; when they appear, weights explode; estimates become unstable

Rejection Sampling vs Importance Sampling

Rejection Sampling

- Accepted samples are exact samples from the posterior

Pros:

- Conceptually simple
- Produces unbiased samples from the true posterior

Cons:

- Requires a global constant M such that $\tilde{p}(\theta) \leq Mq(\theta)$
- Often reject most samples
- Becomes impractical in high dimensions

Importance Sampling

- Approximate expectations by weighted samples

Pros:

- Uses all samples (no rejection)
- No need for a global envelope constant M
- Often more efficient than rejection sampling

Cons

- Performance depends heavily on how well $q(\theta)$ matches p
- Can suffer from weight degeneracy
- High variance in high dimensions

Part III: Core Markov Chain Monte Carlo (MCMC) Algorithms

Why Markov Chains

Goal

- In Bayesian inference, we want to compute expectations under the posterior: $E_{p(\theta|\mathcal{D})}[f(\theta)]$
- **Sampling-based methods** approximate these expectations using samples: $\theta^{(1)}, \dots, \theta^{(N)} \sim p(\theta \mid \mathcal{D})$

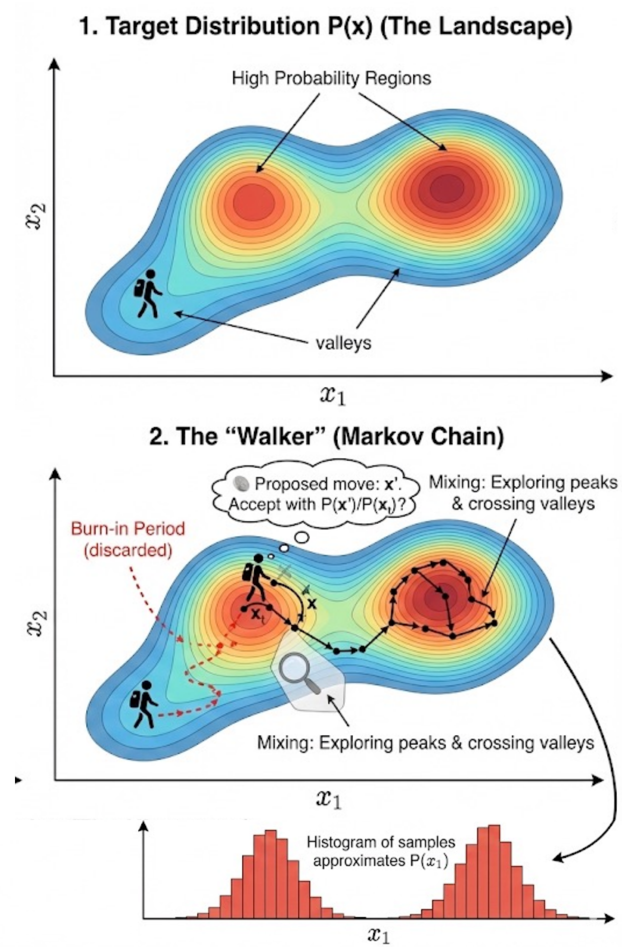
Why not rejection or importance sampling?

- Rejection and importance sampling rely on a **fixed global proposal** $q(\theta)$
- Both suffer high-dimensional spaces:

Key idea

- Instead of proposing samples *from scratch*, move locally and adaptively from the current sample.

Illustration of MCMC dynamics



3. Resulting Samples (The Histogram)

What is a Markov Chain

Intuition: A Markov chain is a dependent sequence of random variables where the next state depends only on the current state.

How the chain evolves

- Let $\theta^{(0)}, \theta^{(1)}, \theta^{(2)}, \ldots,$ be a sequence of random variables (states).
- This sequence forms a first-order Markov chain if:

$$p\left(\theta^{(m+1)} \mid \theta^{(0)}, \ldots, \theta^{(m)} \right) = p\left(\theta^{(m+1)} \mid \theta^{(m)} \right)$$

- We specify transitions using a **transition kernel**: $\theta^{(m+1)} \sim T\left(\cdot \mid \theta^{(m)} \right)$

Components of a Markov chain

- State space: all possible values of θ (In Bayesian inference: the parameter space)
- Initial distribution: $p\left(\theta^{(0)}\right)$
- Transition distribution (kernel): $T(\theta' \mid \theta) = p\left(\theta^{(m+1)} = \theta' \mid \theta^{(m)} = \theta \right)$

Markov Chain for MCMC

Goal of MCMC

We want to generate samples whose long-run distribution is the posterior: $p^*(\theta) \equiv p(\theta \mid \mathcal{D})$

Stationary (invariant) distribution

- A distribution $p^*(\theta')$ is **stationary** for a transition kernel T if:

$$p^*(\theta') = \sum_{\theta} p^*(\theta) T(\theta' \mid \theta)$$

- If $\theta^{(m)} \sim p^*(\theta)$, then after one transition $\theta^{(m+1)} \sim p^*(\theta)$
- The distribution does **not change** as the chain evolves

Why this matters for MCMC

- We often cannot sample directly from $p(\theta \mid \mathcal{D})$
- So we construct a Markov chain whose stationary distribution is $p^*(\theta')$, after enough steps, $\theta^{(0)} \to \theta^{(1)} \to \theta^{(2)} \to \cdots$ the samples behave like draws from $p(\theta \mid \mathcal{D})$

Ergodicity + Detailed Balance

Ergodicity (why "long-run" works)

- A Markov chain is **ergodic** if it eventually *forgets its starting point*.
- In practice: for large m, the distribution of $\theta^{(m)}$ becomes close to $p^*(\theta)$:

$$p\left(\theta^{(m)} \mid \theta^{(0)} \right) \longrightarrow p^*(\theta) \quad \text{as } m \to \infty.$$

- **Burn-in:** early samples depend strongly on $\theta^{(0)}$
- Long-run samples depend only on the stationary distribution to p^*.

Detailed balance (a design rule that guarantees stationarity)

- A sufficient condition for p^* to be stationary is detailed balance:

$$p^*(\theta)T(\theta' \mid \theta) = p^*(\theta')T(\theta \mid \theta')$$

- Probability flow from $\theta \to \theta'$ is equal to flow from $\theta' \to \theta$, so the distribution does not change over time

The Metropolis-Hastings algorithm

Key question: how do we design a transition rule $T(\theta' \mid \theta)$ so that $p^*(\theta) \approx p(\theta \mid \mathcal{D})$ is the stationary distribution?

Challenges so far:

- We want samples from $p(\theta \mid \mathcal{D})$,but direct sampling is impossible
- Transition rule must: be easy to simulate; satisfy detailed balance; explore the space efficiently

Key idea: accept–reject moves inside a Markov chain

- Propose a *candidate move* $\theta' \sim q(\theta' \mid \theta)$
- Accept or reject the move using a probability that corrects for: the proposal distribution and the target posterior

Instead of rejecting samples *globally* (rejection sampling), **Metropolis–Hastings** algorithm reject or accept *local moves* along a Markov chain.

The Metropolis-Hastings algorithm

Goal: construct a Markov chain whose stationary distribution is

$$p(\theta \mid \mathcal{D}) \propto \tilde{p}(\theta) = p(\mathcal{D} \mid \theta)p(\theta)$$

Algorithm (single step)

Given current state $\theta^{(m)}$:

1. Propose a move

$$\theta' \sim q\left(\theta' \mid \theta^{(m)}\right)$$

2. Compute acceptance probability

$$\alpha = \min\left(1, \frac{\tilde{p}(\theta')q\left(\theta^{(m)} \mid \theta'\right)}{\tilde{p}(\theta^{(m)})q\left(\theta' \mid \theta^{(m)}\right)}\right)$$

3. Accept or reject

- Sample $u \sim \mathrm{Uniform}(0,1)$
- If $u \leq \alpha$: set $\theta^{(m+1)} = \theta'$
- Else: set $\theta^{(m+1)} = \theta^{(m)}$

The Metropolis algorithm

The **Metropolis algorithm** is a special case of MH algorithm where the proposal distribution is symmetric.

Symmetric proposal

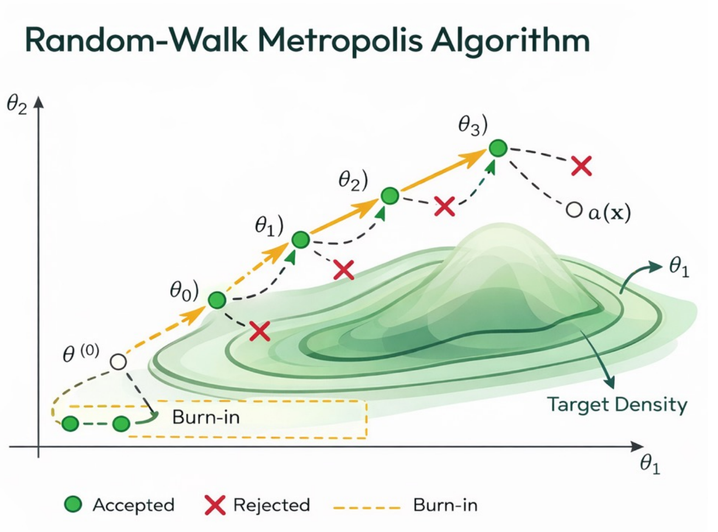
- We choose a proposal distribution such that: $q(\theta' \mid \theta) = q(\theta \mid \theta')$
- A common choice: $q(\theta' \mid \theta) = \mathcal{N}(\theta, \Sigma)$ (Random-walk Gaussian)

Acceptance probability simplifies

$$\alpha\left(\theta^{(m)}, \theta'\right) = \min\left(1, \frac{\tilde{p}(\theta')}{\tilde{p}(\theta^{(m)})}\right)$$

Why Metropolis is useful

- Very simple to implement
- Requires only the unnormalised posterior
- Forms the foundation of many MCMC methods



Gibbs Sampling

Core idea:

- Instead of proposing and rejecting, we sample directly from the conditional distributions:

$$\theta_k^{(m+1)} \sim p(\theta_k \mid \theta_{-k}^{(m)}, \mathcal{D}), \text{ where } \theta = (\theta_1, \dots, \theta_K) \text{ and } \theta_{-\mathrm{k}} = \text{ all parameters except } \theta_{\mathrm{k}}$$

- We update one component at a time.

Gibbs a Special Case of MH

- In Gibbs sampling, the proposal is chosen as:

$$q\left(\theta_k' \mid \theta_{-k}^{(m)}\right) = p\left(\theta_k \mid \theta_{-k}^{(m)}, \mathcal{D}\right),$$

- When we plug this into the MH ratio, everything cancels and: $\alpha = 1$

Interpretation

- Every proposal is accepted
- No rejection step
- Moves are always valid under the posterior

Gibbs Sampling

University of Exeter

Algorithm (block updates)

Goal: sample from the posterior $p(\theta \mid \mathcal{D})$,where $\theta = (\theta_1, \dots, \theta_K)$

- For iteration m:

$$\theta_1^{(m+1)} \sim p(\theta_1 \mid \theta_2^{(m)}, \dots, \theta_K^{(m)}, \mathcal{D})$$

$$\theta_2^{(m+1)} \sim p(\theta_2 \mid \theta_1^{(m)}, \dots, \theta_K^{(m)}, \mathcal{D})$$

$$\vdots$$

$$\theta_K^{(m+1)} \sim p(\theta_K \mid \theta_1^{(m)}, \dots, \theta_K^{(m)}, \mathcal{D})$$

Cycle through all components

When Does Gibbs Work Well?

- Each conditional $p(\theta_k \mid \theta_{-k}^{(m)}, \mathcal{D})$, is easy to sample from (often conjugate models)
- Parameters are not too strongly correlated (mixing is faster)
- Conditionals have wide enough support (chain can move around)

Summary

Inference = Integration

- Bayesian inference requires computing expectations: $E_{p(\theta|\mathcal{D})}[f(\theta)]$
- Monte Carlo approximates integrals using **sample averages**

Direct Monte Carlo Methods

- **Rejection sampling** → exact but inefficient in high dimensions
- **Importance sampling** → uses all samples but can suffer from weight degeneracy
- Both rely on a **fixed proposal distribution**

Why MCMC?

- Fixed global proposals fail in high dimensions
- Instead, construct a **Markov chain** $\theta^{(0)} \to \theta^{(1)} \to \cdots$
- Designed so its **stationary distribution** is: $p(\theta \mid \mathcal{D})$

Core MCMC Algorithms

- **MH** = general accept/reject framework
- **Metropolis** = MH with symmetric proposal case
- **Gibbs** = MH where the proposal is the exact conditional, so acceptance is always 1